

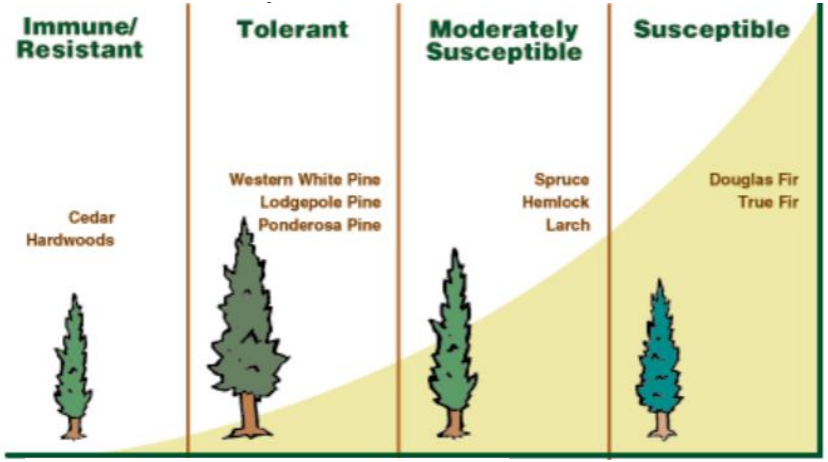
Thinning, hardwood rotation, and laminated root rot in Douglas fir

David Showalter, Shawn McMurtrey, Anna Leon, John Browning, Michael McWilliams, Blakey Lockman, Everett Hansen, Jared LeBoldus

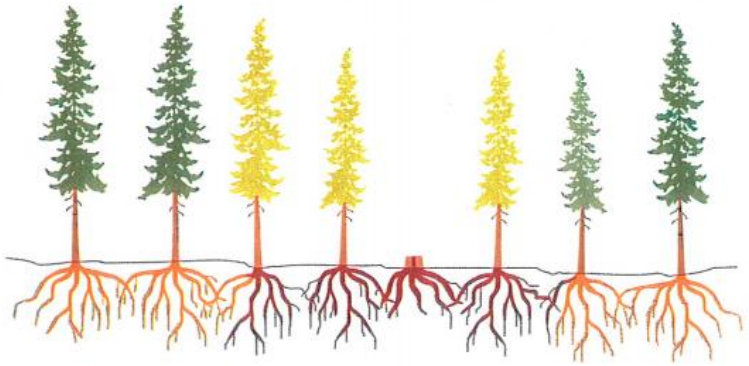
- *Coniferiporia sulphurascens* (Basidiomycota:Agaricomycetes)
- Several coniferous hosts
- Both saprophytic and parasitic
- Spread through root to root contact between trees and/or stumps
- Disease of the site, for 50 years (Hansen, 1976)
-



Kevin Zobrist



Dr. Richard Hamelin & Dr. Allan Carroll



Sturrock & Garbutt, 1994

Signs and Symptoms



Near Waldo Lake, OR

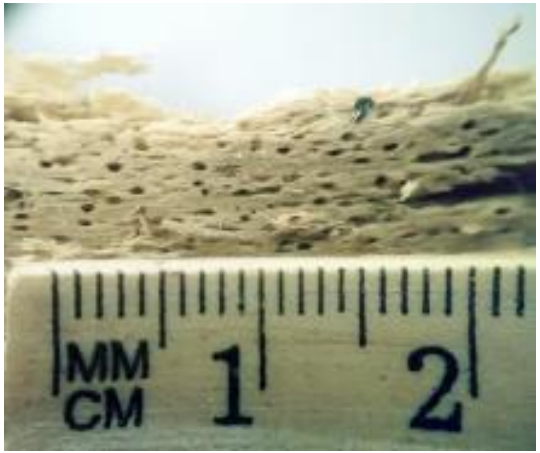




Kevin Zobrist



David Shaw, pnwhandbooks.org

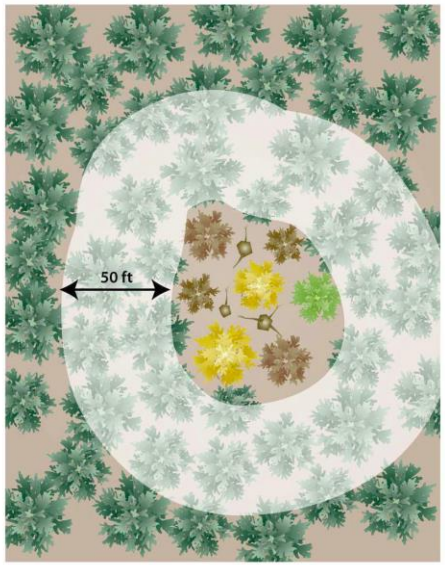


Impacts

- Most damaging root rot of *Pseudotsuga menziesii* in PNW.
- Infected sites result in a 40% - 70% reduction in wood volume
- Oregon timber industry, $12 billion annually, over 58,000 jobs

Management

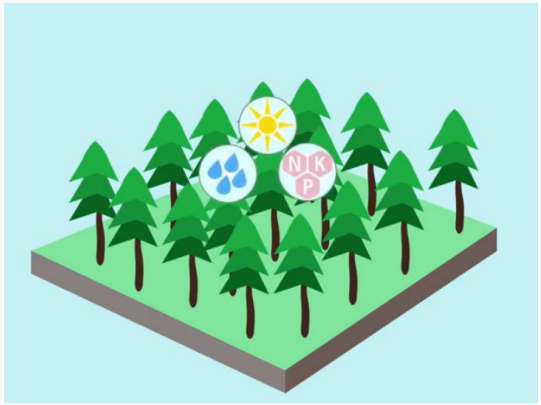
- Buffer cutting, thinning
- Inoculum removal (stumping)
- Rotating less suceptible host



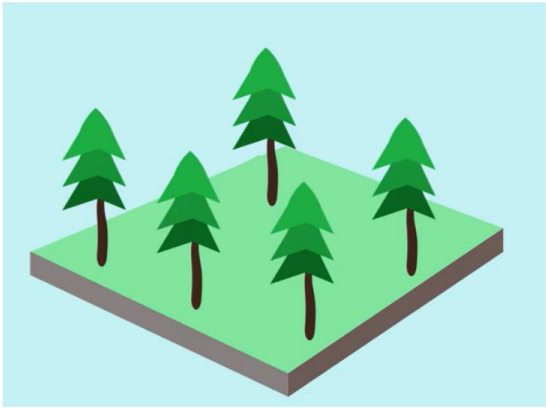


Thinning

- Benefits the local economy and wildlife.
- Help prevent fire spreading to crowns of larger trees.
- Increase health and reduces some disease threats.
- Improve vegetation diversity of the forest floor.
- Increase tree growth and vigor of remaining trees.







Source and illustrations: US Forest Service, 2020

Research Questions

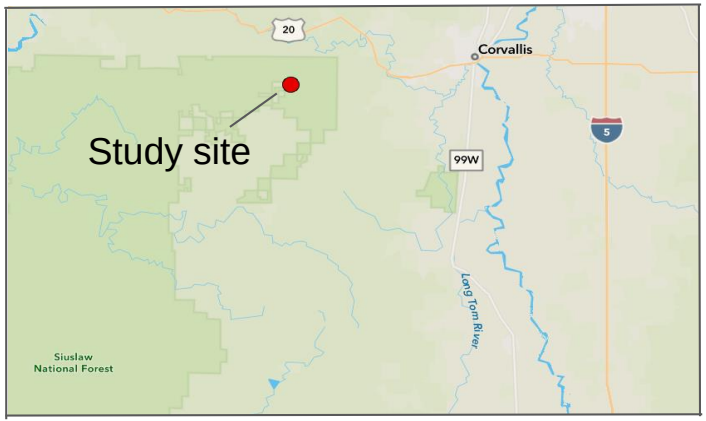
1. Do thinning treatments in mature stands with high incidence of laminated root rot increase tree growth and/or reduce mortality?

2. Do rotations of non-host hardwood species in areas with high LRR incidence increase growth and reduce mortality in subsequent Douglas fir plantings?

3. How well do the Forest Vegetation Similator and its Western Root Disease Model variant predict long term growth in these stands?

Siuslaw National Forest

- Unique ecosystem for surveying of LRR.
- Study area was clear cut between 1910 and 1920.
- Area burned after reharvest, reestablished from seed (Tkacz & Hansen, 1982).
- Study area is ~400 acres.



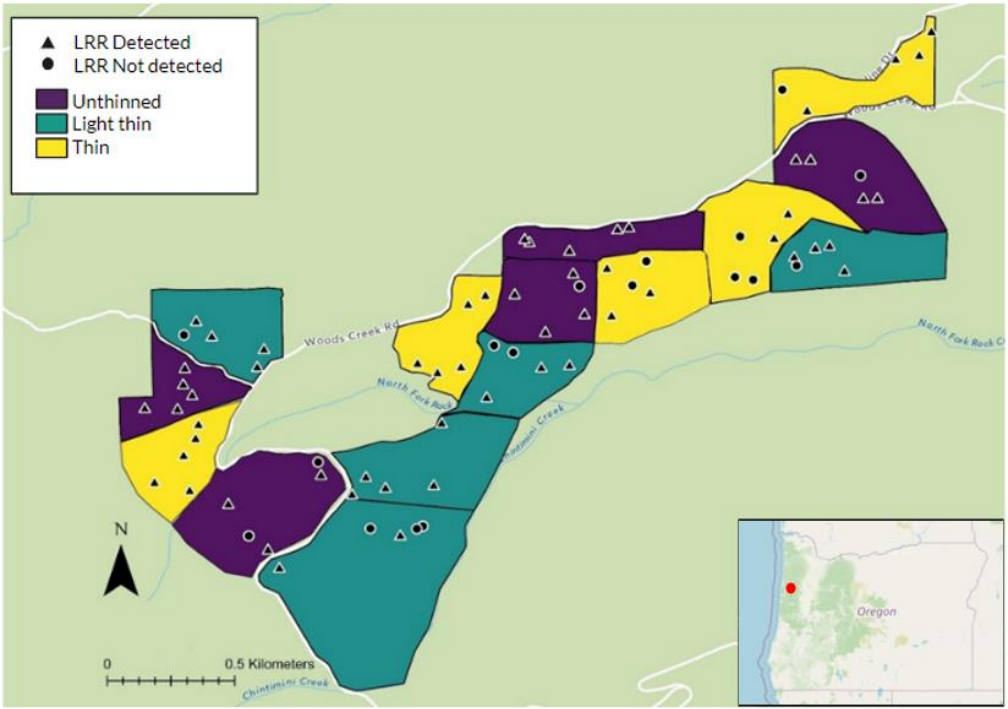


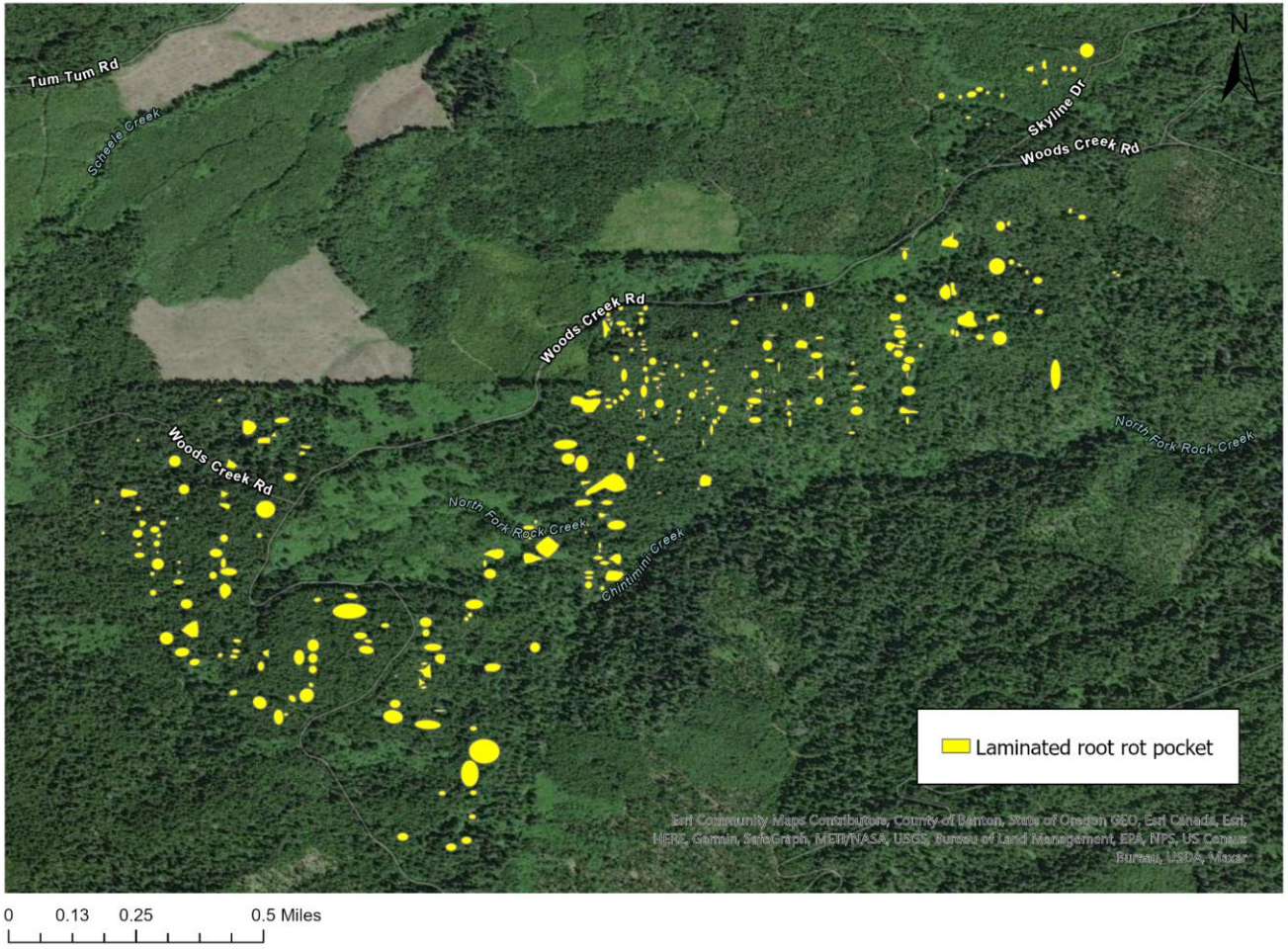
Adam Carson

Woods Creek Study Area

- Mature mixed Douglas fir forest, ~65y old at establishment in 1985. High LRR incidence
- Network of 15 stands and 75 plots over 162 hectares
- Thinning treatments applied to stands ranging from ~6-14 hectares.
- Circular 1/25 hectare sample plot , 11.3 m radius.
- Measured aprox. every 5-10 years
- Data recorded was **species,** diameter at breast height (**DBH**), and **tree health condition.**



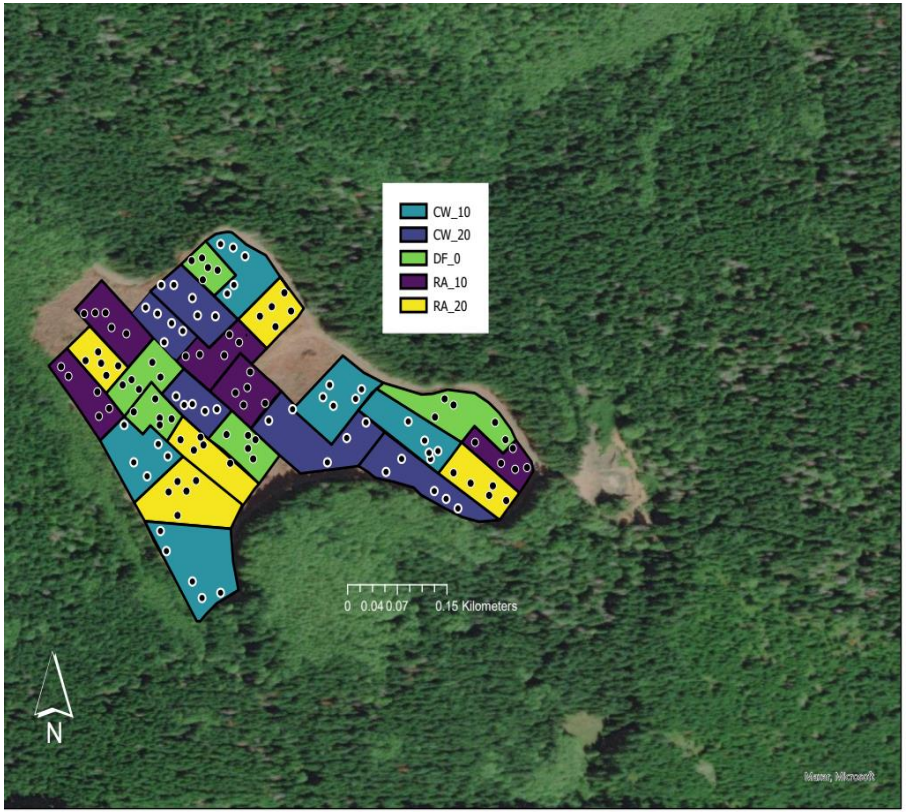


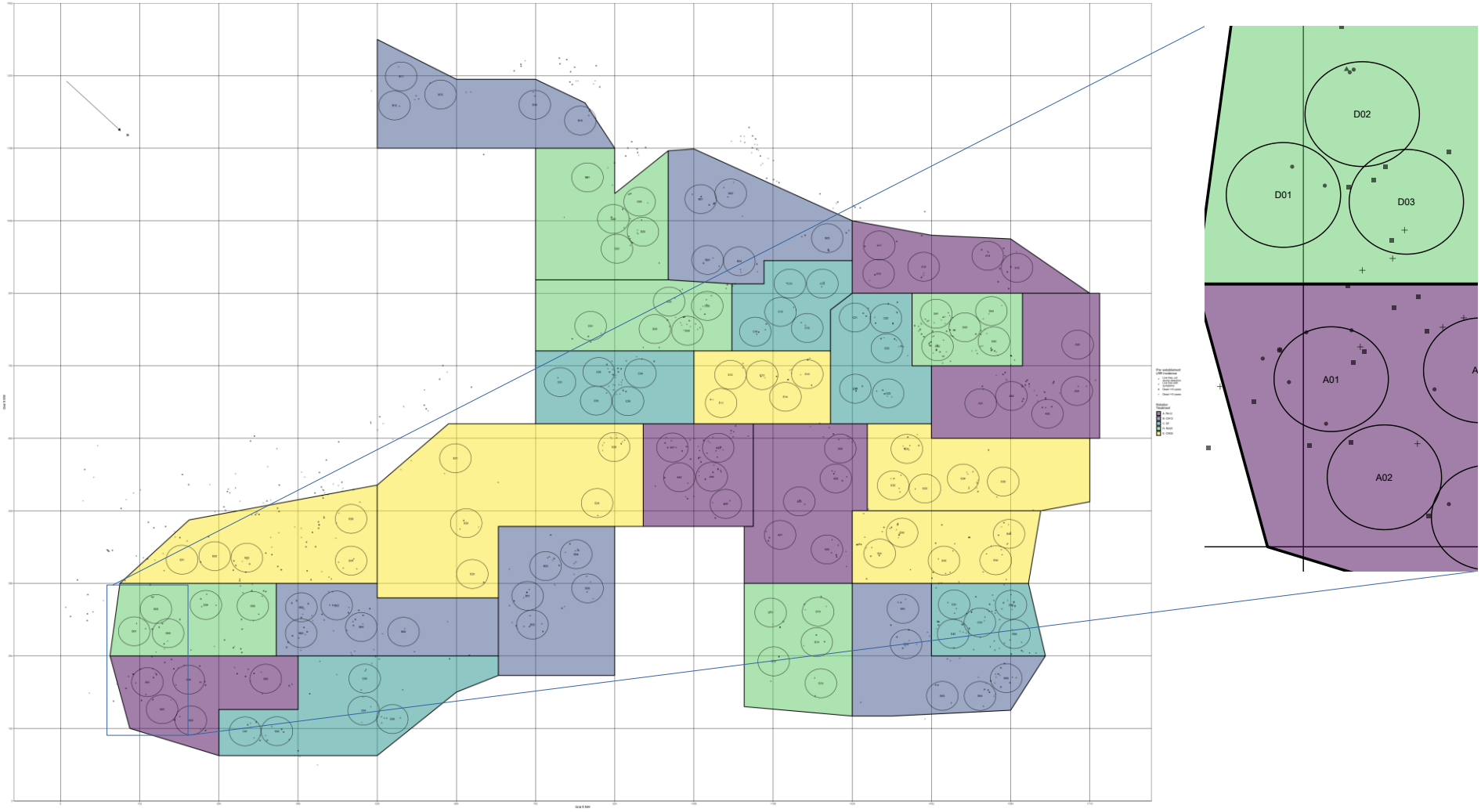


(Hansen, 1978. Line intercept survey. Bloomberg et al., 1980.)

Apiary Study Area

- Newly planted Douglas fir following clearcut
  of mixed Douglas fir forest
  with high LRR incidence
- Network of 25 stands and 125 plots
  over 14 hectares in Columbia County
  in far NW Oregon
- Roation treatments applied to stands
- Stands blocked by initial LRR incidence
- Circular 1/50 hectare sample plot ,
  8 m radius.
- Measured sporatically, every 8-24 years
- Data recorded was diameter at
  breast height (**DBH**), and **tree health condition.**



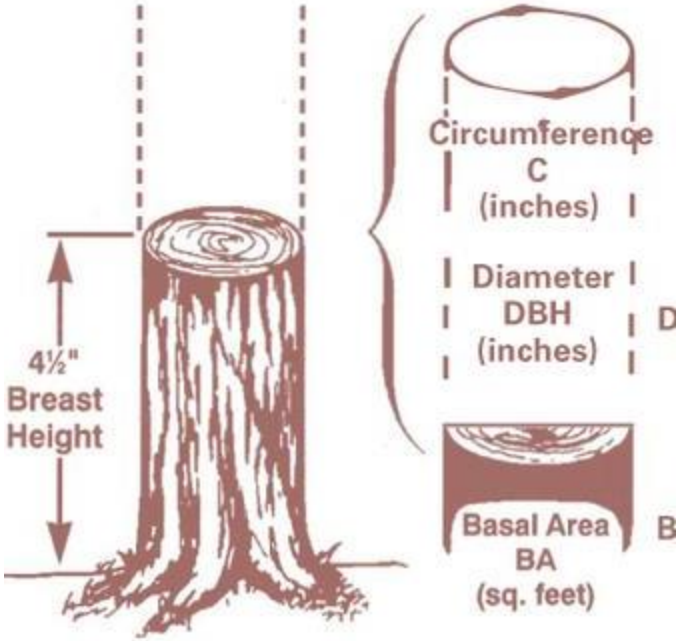

D02
D01
D03
A01
A02

Forestry Biometrics 101

Diameter at breast height (**DBH**)
(cm or in)
1-dimesional ~ distance

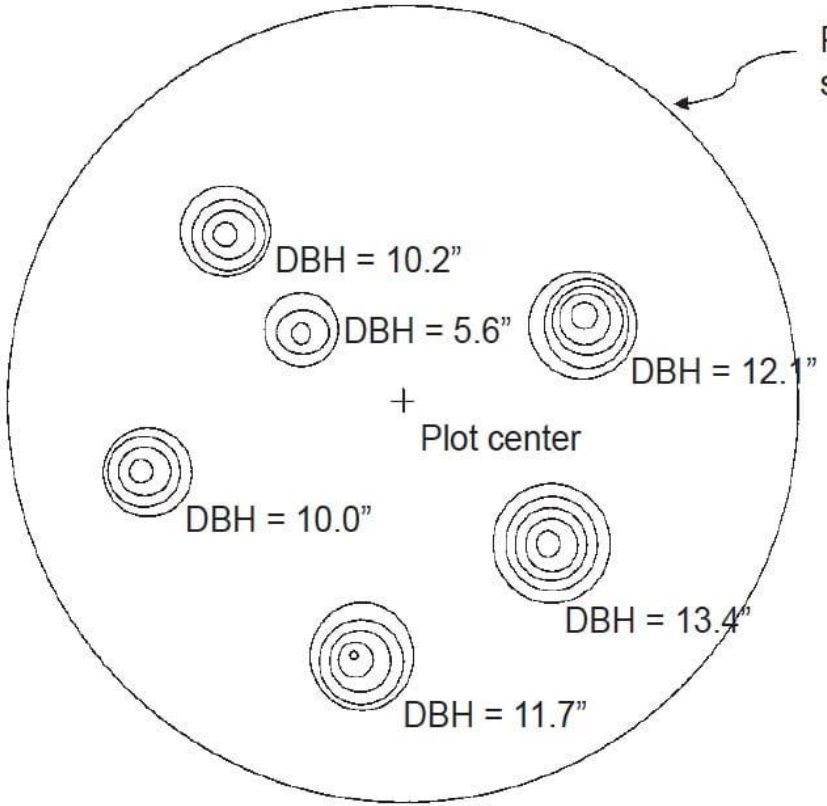


Tree basal area (**BA**)
(m^2 or ft^2)
2-D ~ area



Stand basal area (**BA**)
(m^2/hectare or ft^2/acre)
2-D ~ density

Stand > plot

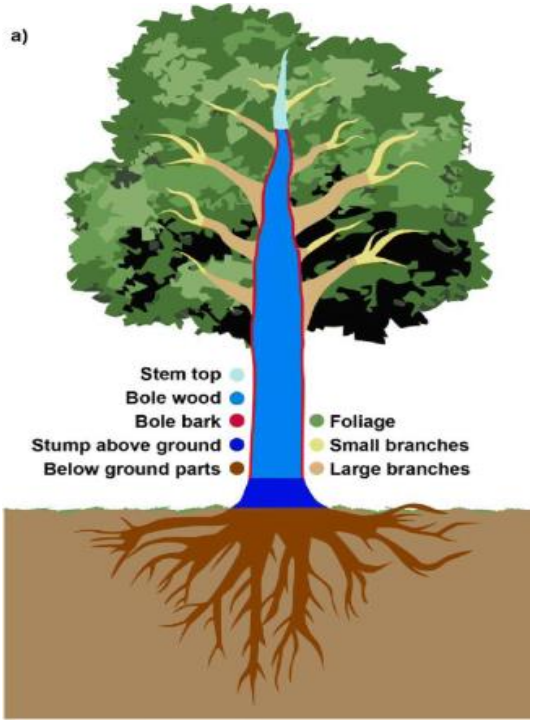


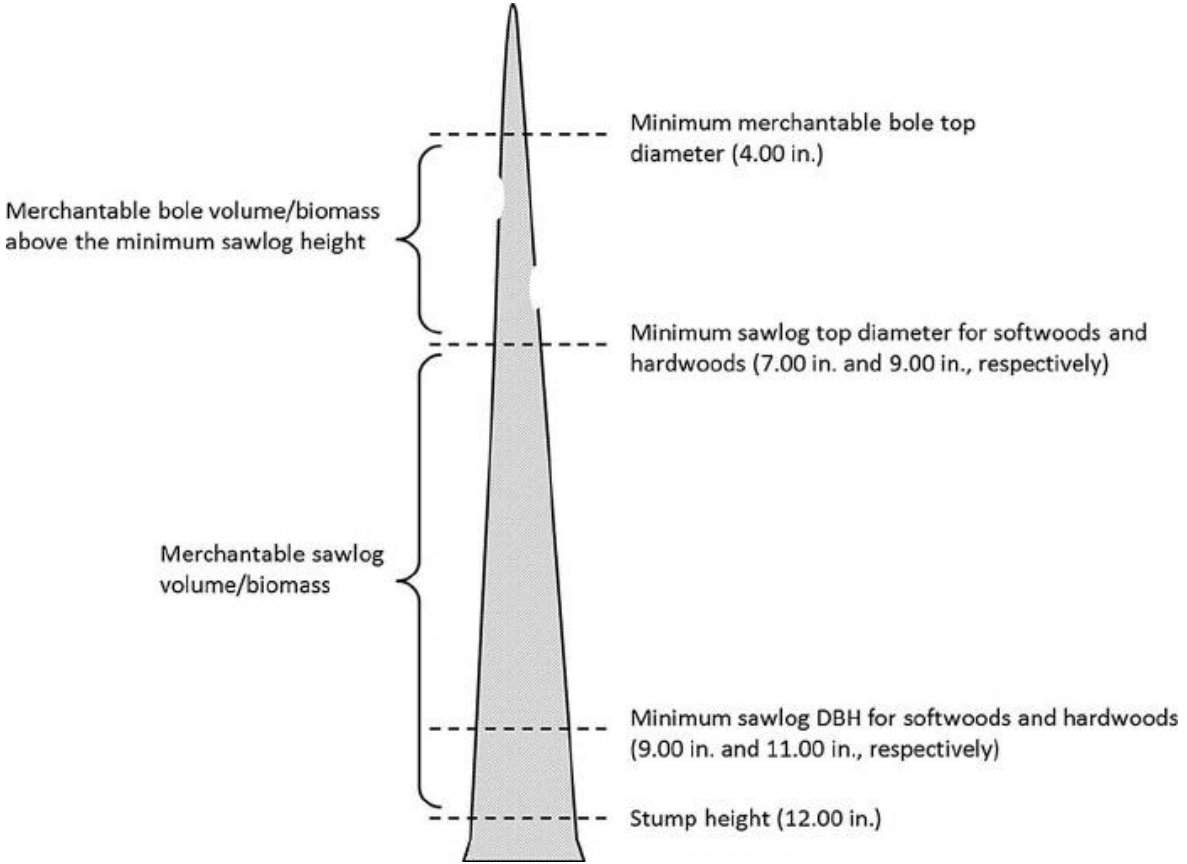
DBH (inches)	Basal Area (square feet)
10.2	0.567
5.6	0.171
12.1	0.799
8.4	0.385
13.4	0.979
11.7	0.747
10.0	0.545
Sum	4.193

Basal area on one 1/10-acre plot = 4.193 sq ft

Basal area per acre (BA/ac) = 4.193 sq ft per 1/10 acre or 41.93 sq ft/ac

Wood Volume
(m^3 or ft^3)
3-D

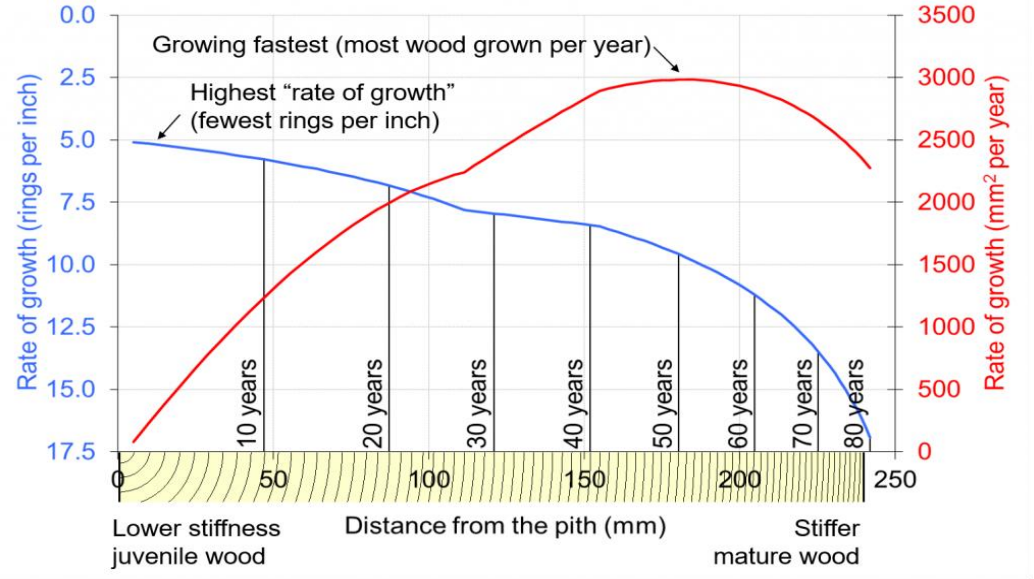




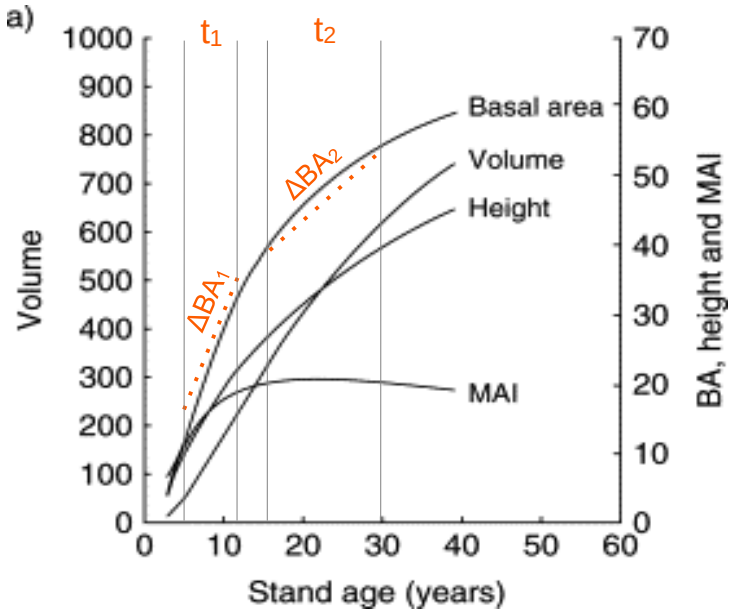
Increment
(metric/time)
rate or slope ~ **growth**

Relationships beween different metrics
(DBH, height, BA, volume, age, increments)
are context dependent, and active areas of research

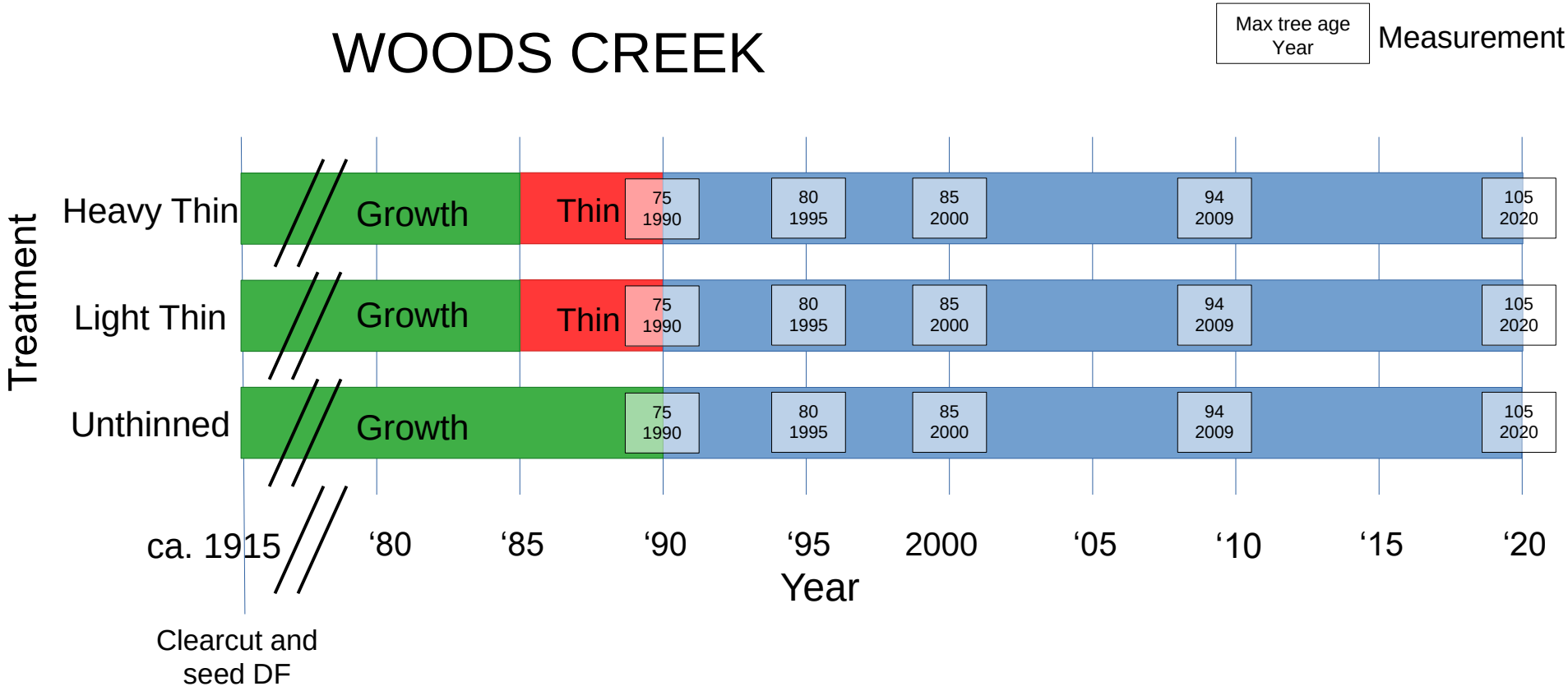
ΔDBH/year



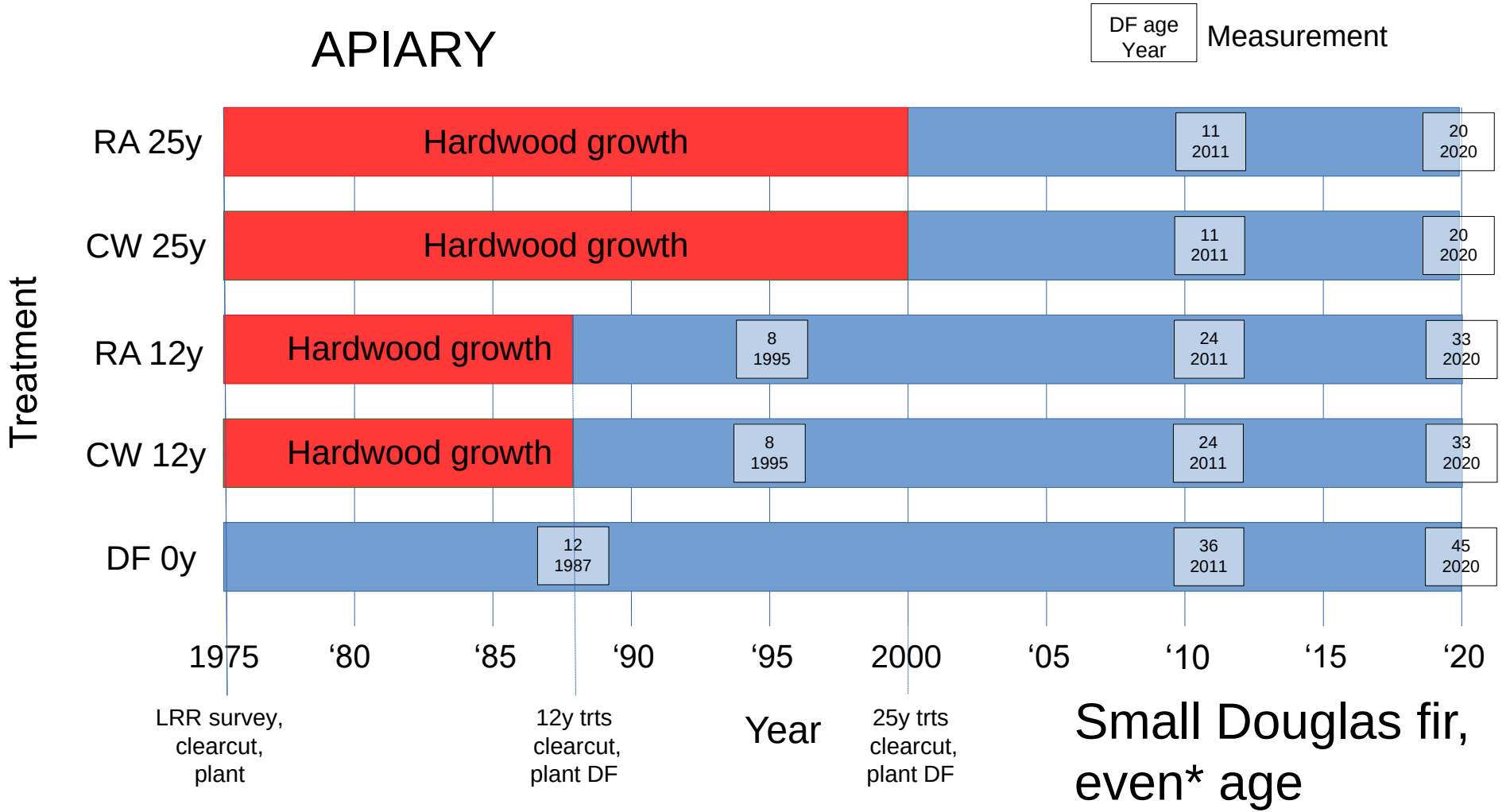
ΔBA/year

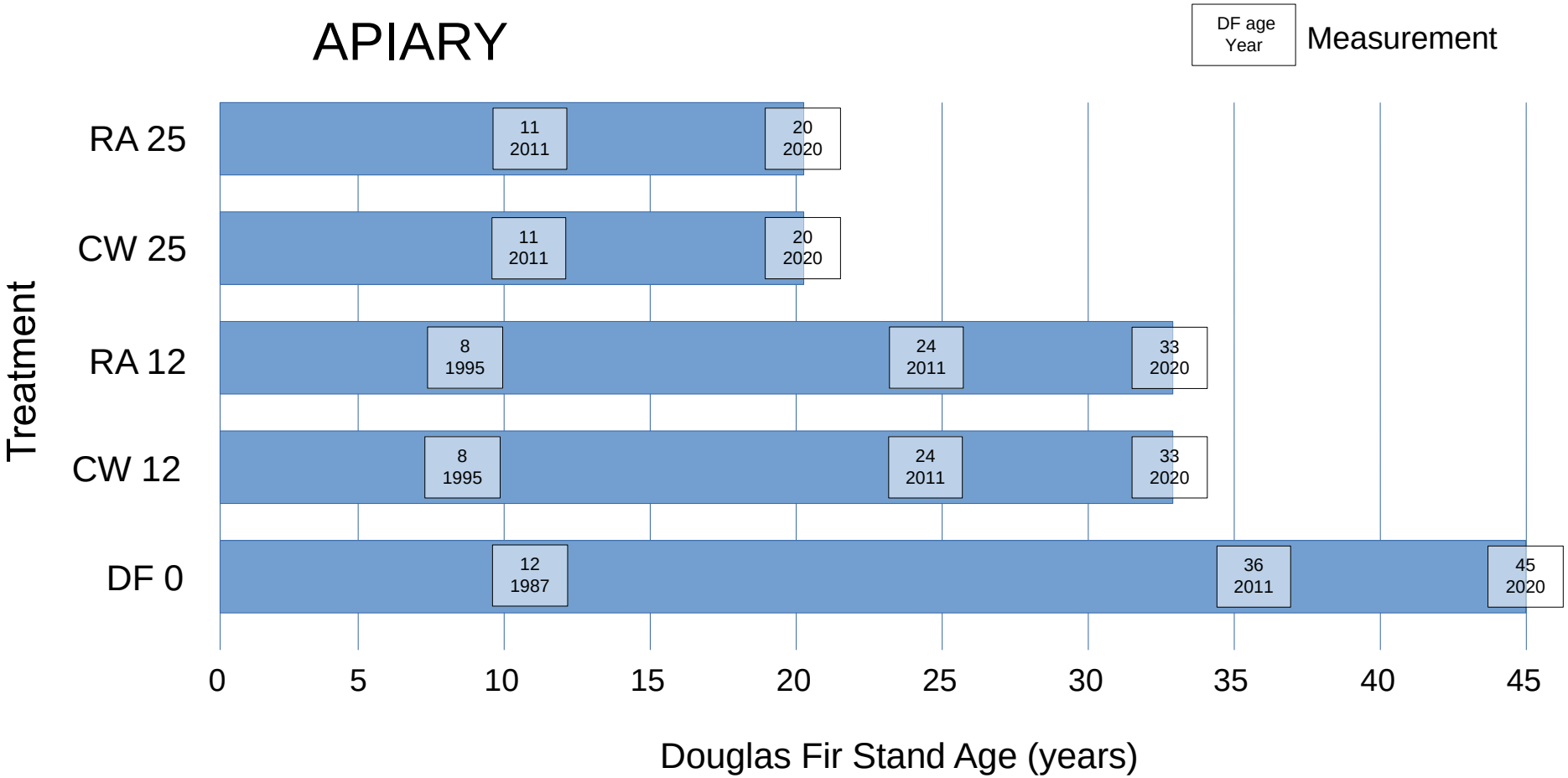


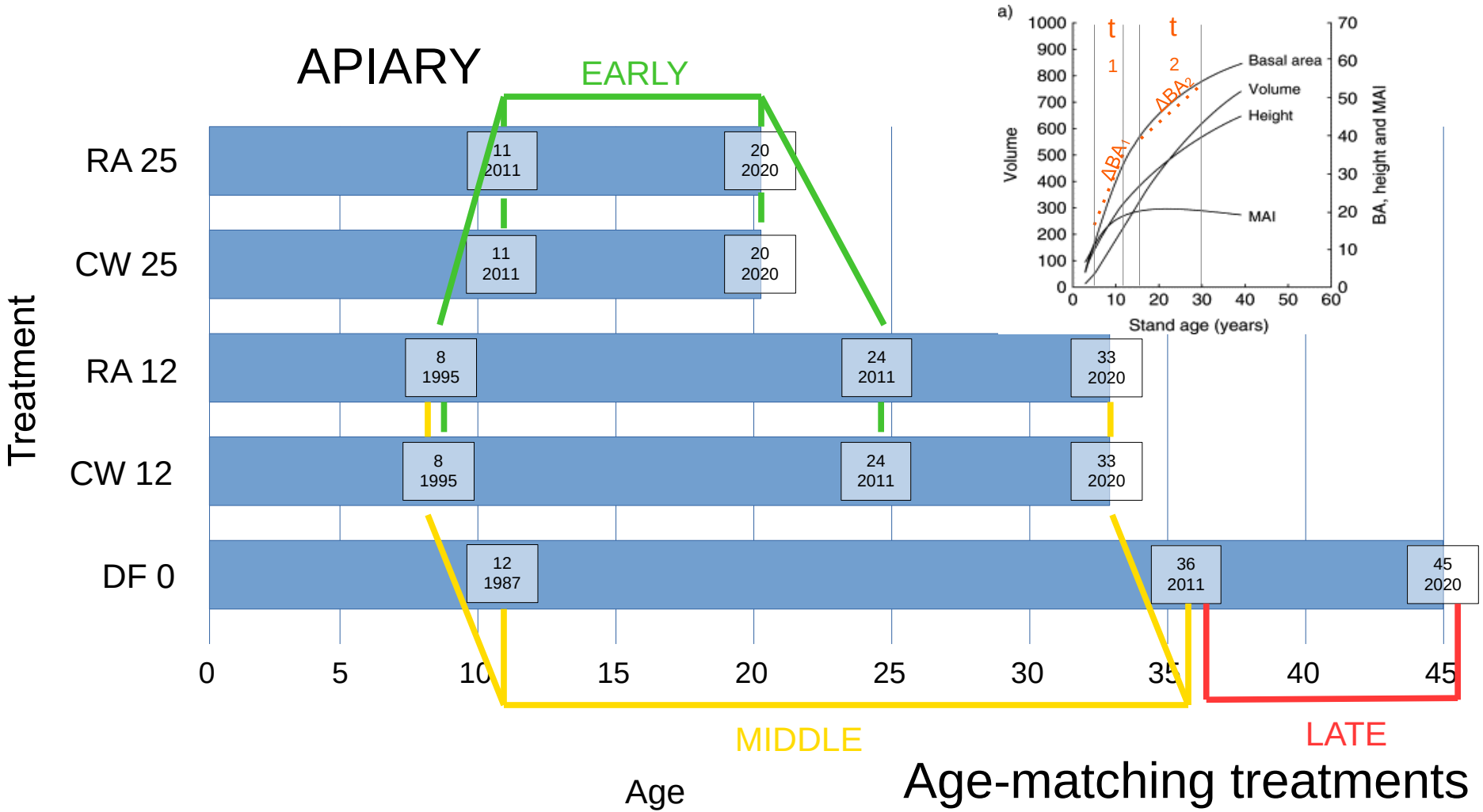
WOODS CREEK



Large old trees, mixed age and species


APIARY
DF age
Year
Measurement
Treatment
RA 25y
CW 25y
RA 12y
CW 12y
DF 0y
Hardwood growth
Hardwood growth
Hardwood growth
Hardwood growth
11
2011
20
2020
11
2011
20
2020
8
1995
24
2011
33
2020
8
1995
24
2011
33
2020
12
1987
36
2011
45
2020
1975
‘80
‘85
‘90
‘95
2000
‘05
‘10
‘15
‘20
Year
LRR survey, clearcut, plant
12y trts clearcut, plant DF
25y trts clearcut, plant DF
Small Douglas fir, even* age


APIARY
DF age
Year
Measurement
Treatment
RA 25
CW 25
RA 12
CW 12
DF 0
11
2011
20
2020
11
2011
20
2020
8
1995
24
2011
33
2020
8
1995
24
2011
33
2020
12
1987
36
2011
45
2020
0
5
10
15
20
25
30
35
40
45
Douglas Fir Stand Age (years)


APIARY
EARLY
RA 25
CW 25
RA 12
CW 12
DF 0
Treatment
11 2011
20 2020
11 2011
20 2020
8 1995
24 2011
33 2020
8 1995
24 2011
33 2020
12 1987
36 2011
45 2020
0
5
10
15
20
25
30
35
40
45
Age
MIDDLE
LATE
Age-matching treatments
a)
t 1
t 2
ΔBA1
ΔBA2
Basal area
Volume
Height
MAI
Volume
BA, height and MAI
Stand age (years)

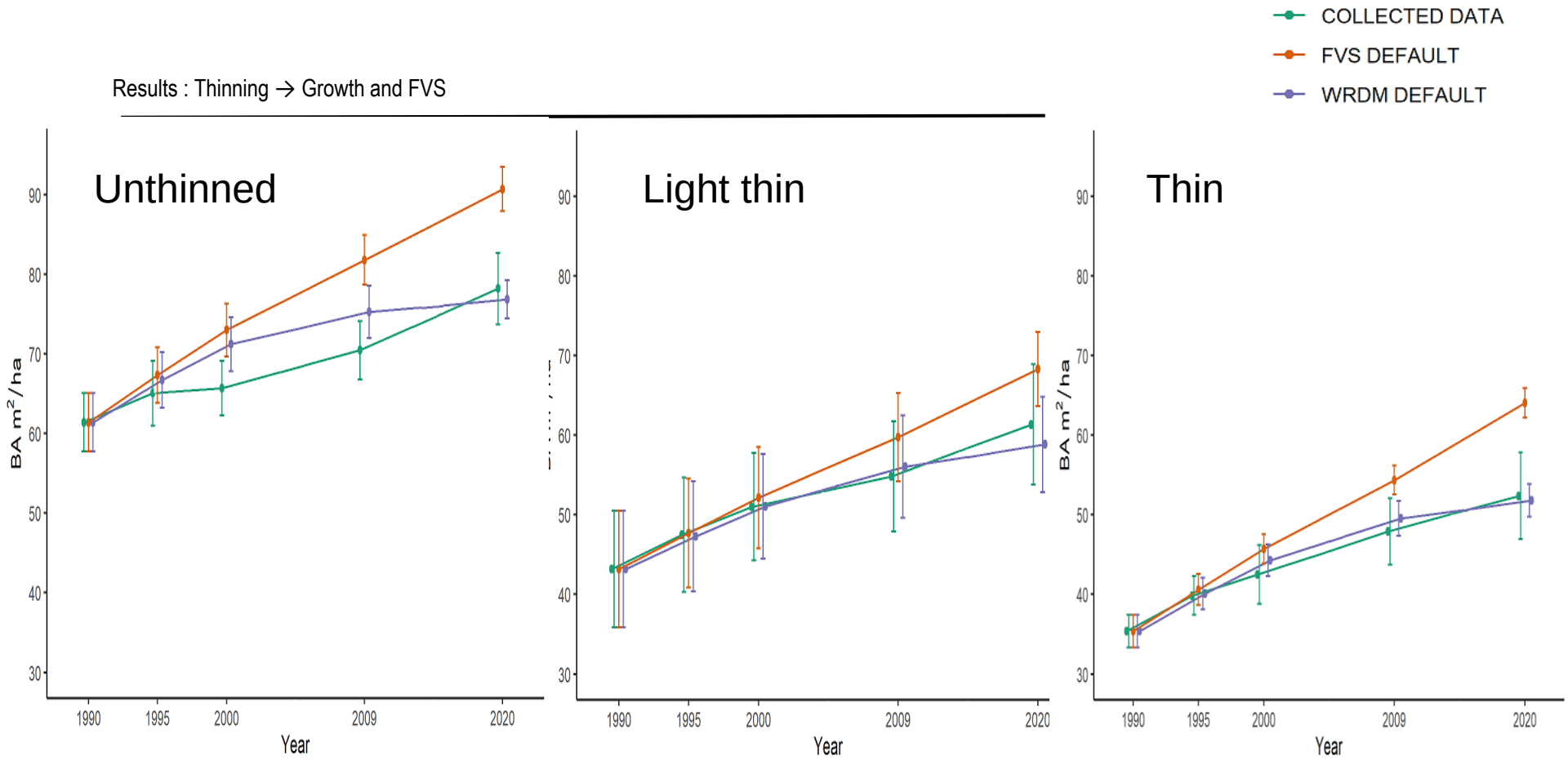
Long term mortality:
(LRR Dead 2020 – LRR Dead 1990)/ Total

Treatment	Net Dead	Total	%
Unthinned	92	361	25.4
Light Thin	72	195	36.9
Thin	60	172	34.8

PLOT over time?
stand mean +/- SE?
Mortalitiy is sparse

$F_{2,12}=0.484$
$p=0.629$

Results : Thinning → Growth and FVS



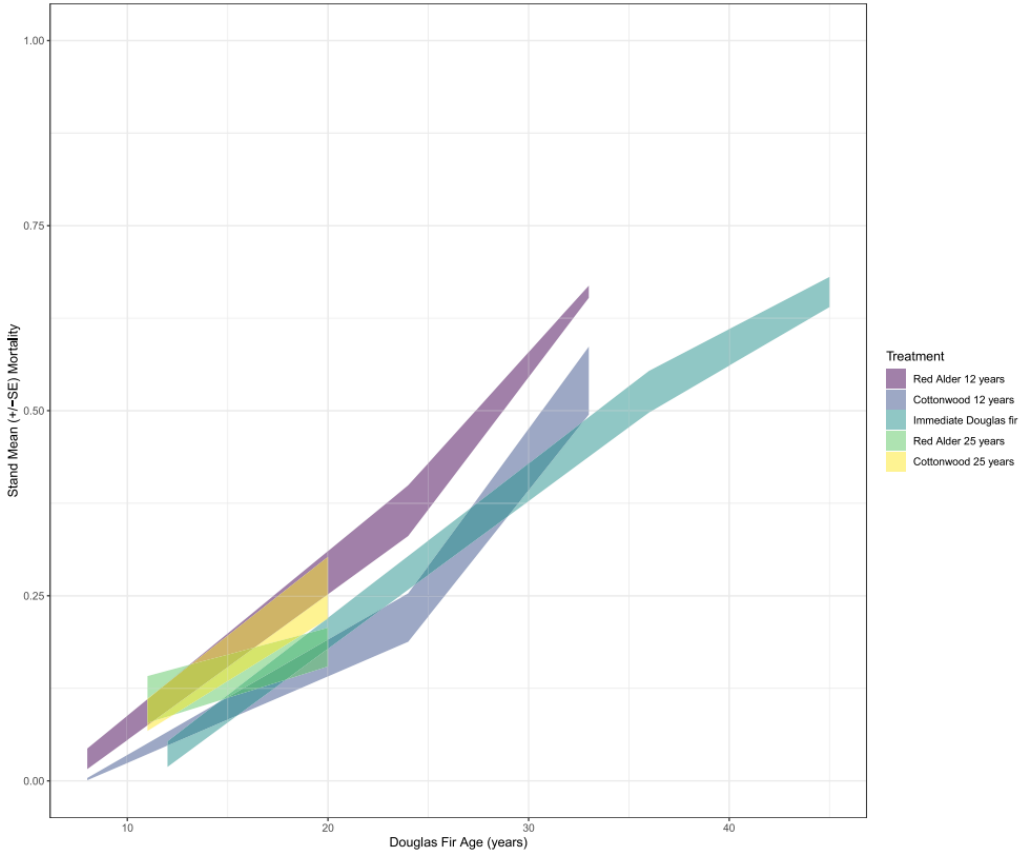
Woods Creek Results : THM

- No observed effect of thinning on long-term rate of mortality
- As yet unknown effect of thinning on basal area increments
- FVS Western Root Disease Model very accurately predicts stand level basal area over the course of the experiment.

Woods Creek Results : Clarifications or Edits

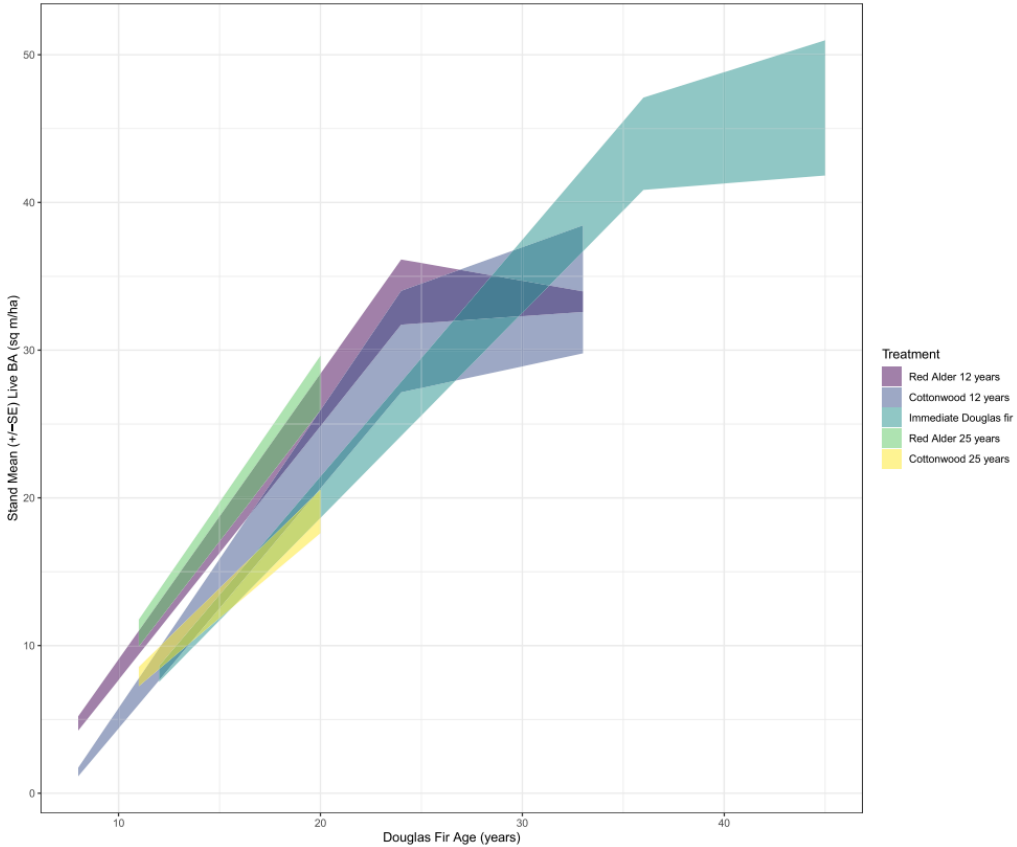
- make a BA x time plot with all treatments
- make a mort x time plot with all treatments?
- Calculate BA annual increments?
- Tree size distribution?
- Growth by LRR status (later)

Results : Rotation → Mortality

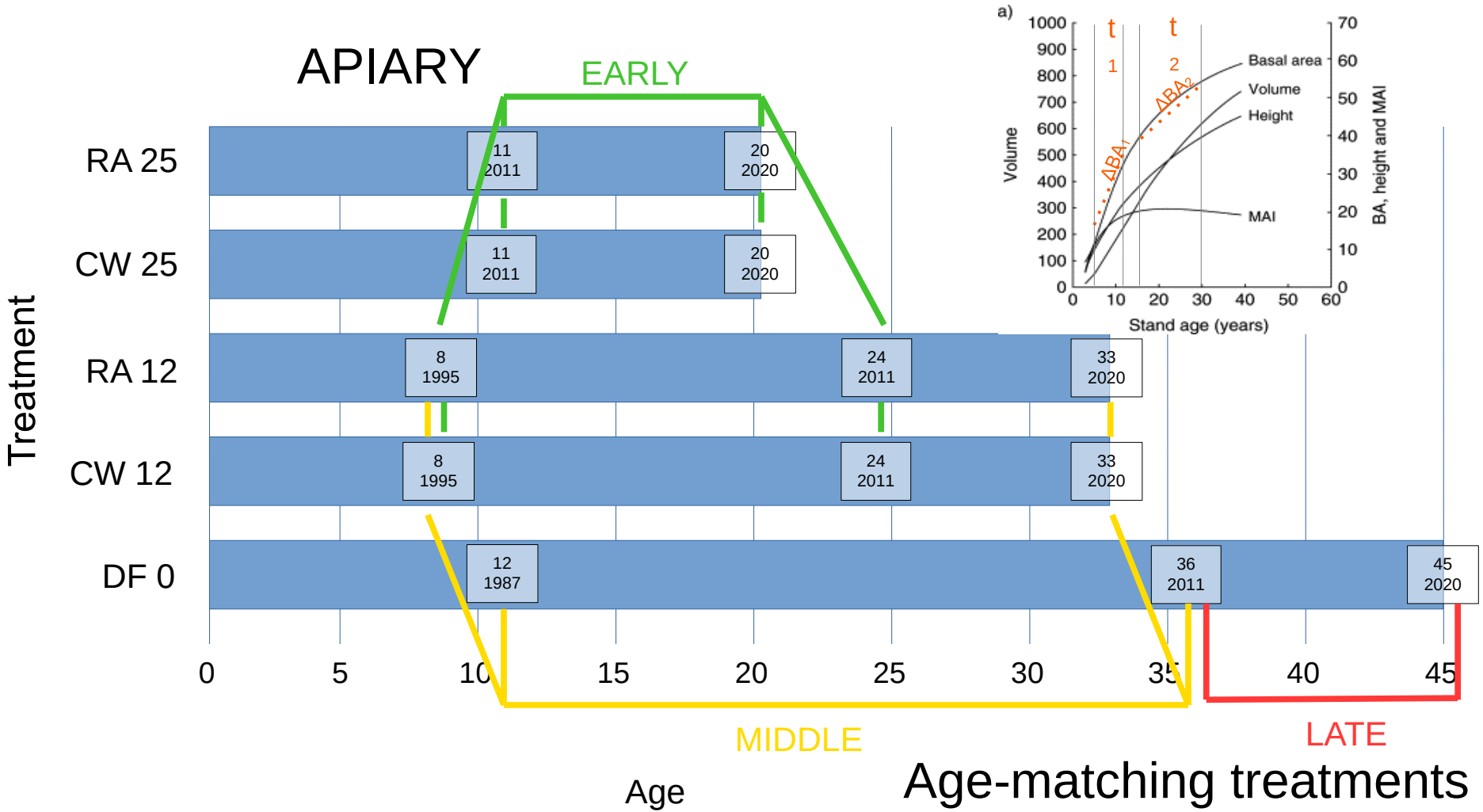


- Mortality increases across the study
- Variation within treatments approximately equal to variation between treatments
- Rate of mortality appears variable

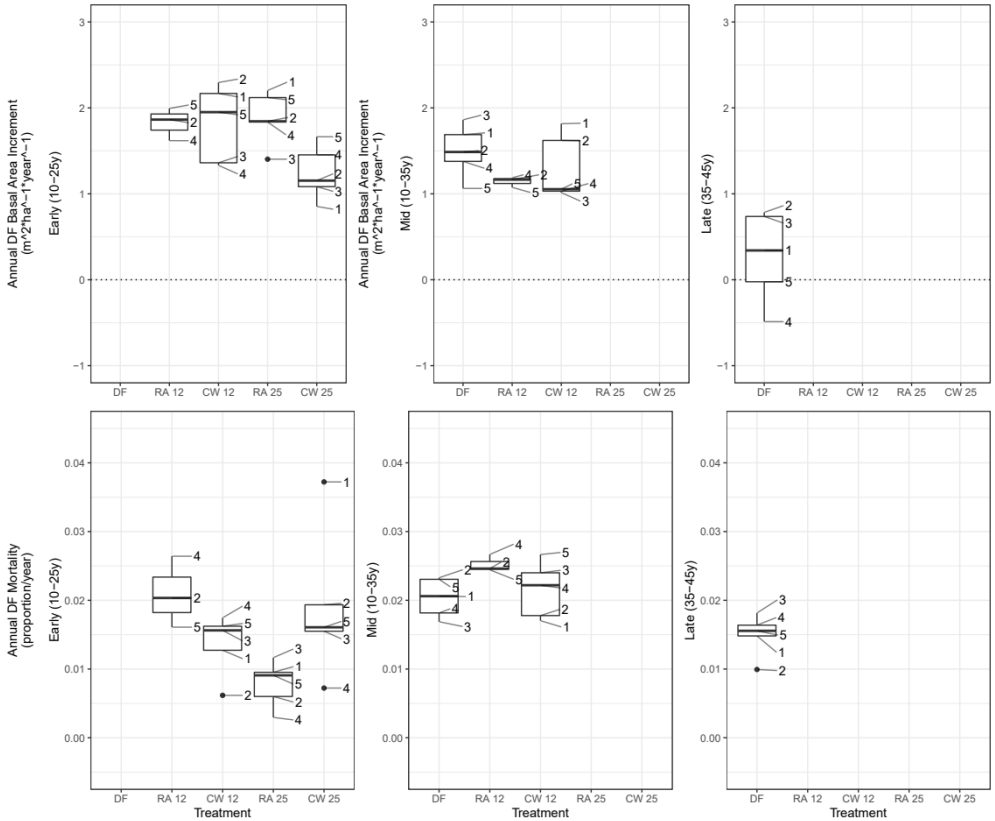
Results : Rotation → Growth



- Growth measured as stand basal area increases sharply early in the study and more slowly later on
- Variation within treatments approximately equal to variation between treatments
- Rate of growth is variable

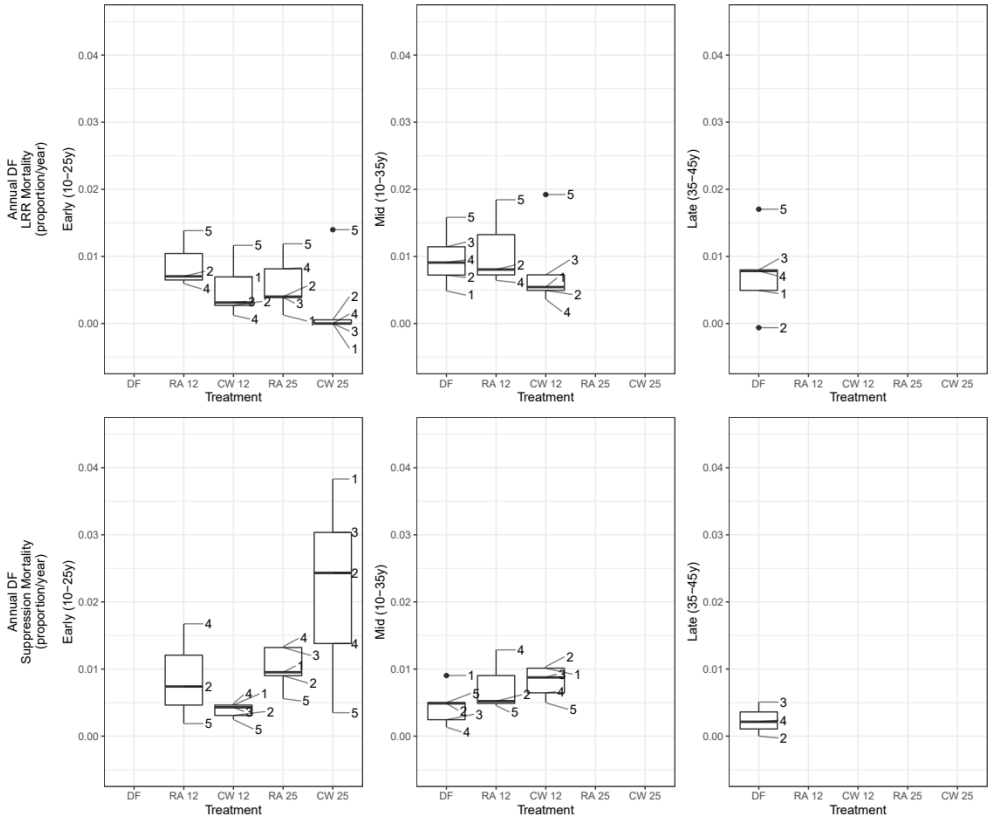

APIARY
EARLY
RA 25
CW 25
RA 12
CW 12
DF 0
Treatment
11 2011
20 2020
11 2011
20 2020
8 1995
24 2011
33 2020
8 1995
24 2011
33 2020
12 1987
36 2011
45 2020
0
5
10
15
20
25
30
35
40
45
Age
MIDDLE
LATE
Age-matching treatments
a)
t 1
t 2
ΔBA1
ΔBA2
Basal area
Volume
Height
MAI
Volume
BA, height and MAI
Stand age (years)

Results : Rotation → Increments



- BA growth increment within timeframes does not significantly vary between treatments
- High LRR severity blocks appear to have reduced growth and increased mortality after early phase
- Growth declines or even becomes negative in later stages, especially in severe blocks

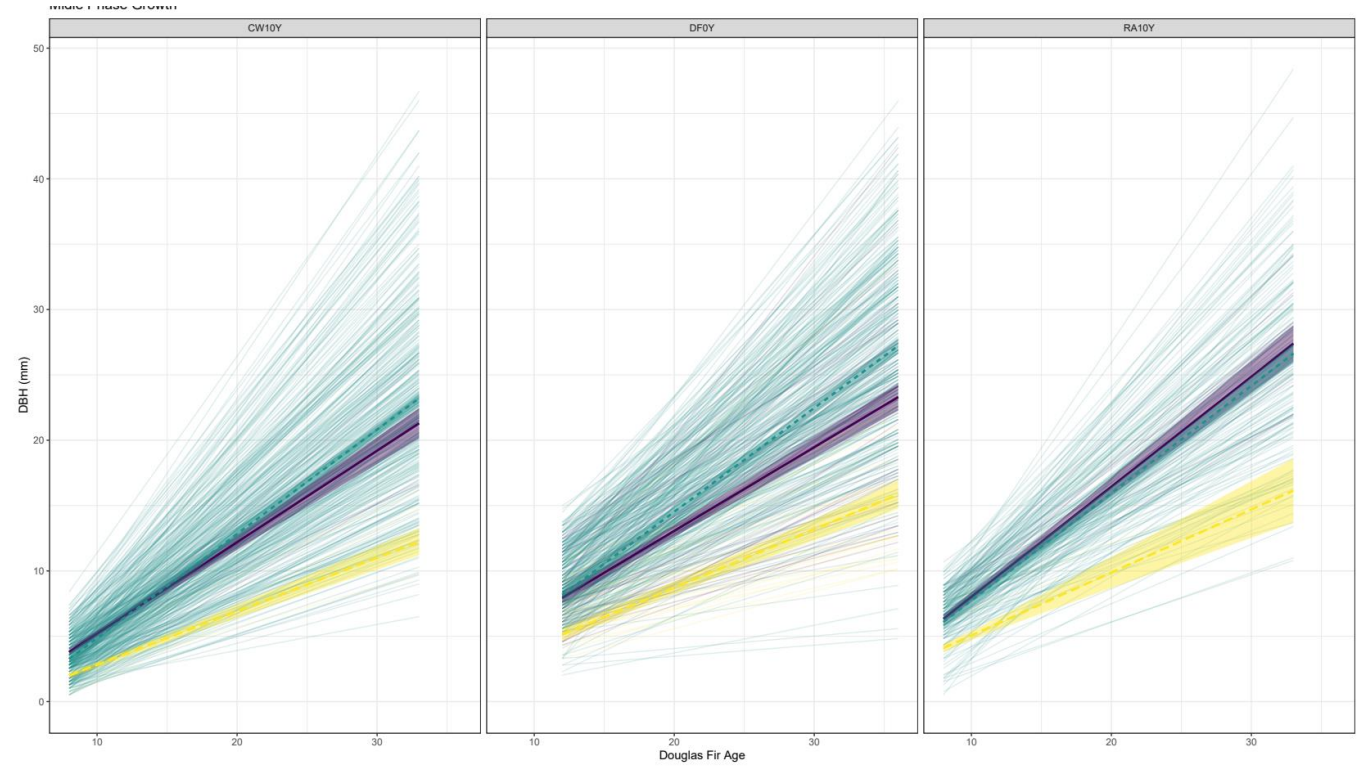
Results : Rotation → Increments by cause

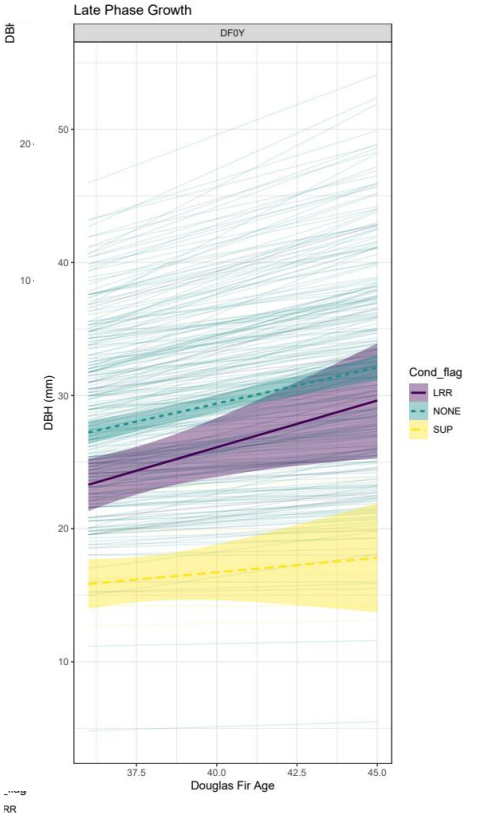


- Suppression by hardwood regeneration may offset LRR mortality reduction
- Initial LRR severity inffluences LRR mortality

Results : LRR → DBH increment

Trees binned “ever SUPP”, “ever LRR”, or “not”. Only living DBH used





Apiary Results : THM

- Suppression by hardwood regeneration may offset LRR mortality reduction, resprout management important
- LRR may have sublethal effects on growth

Apiary Results : Clarifications or Edits

-
-
- Tree size distribution?
- FVS of continuous DF treatment